

Practical Task 2

1. The Python `requests` Module

What is the `requests` module?

The `requests` module is a Python library that makes it easy to communicate with websites and web APIs. It lets a Python program send requests to a server and receive information back. Instead of writing a lot of complicated networking code, the `requests` module gives us simple functions that are easy to use.

How is it used to make HTTP requests?

The `requests` module can send different types of HTTP requests depending on what we want to do.

- **GET** is used to get information from a website or API.
- **POST** is used to send new information to a server.
- **PUT** is used to update existing information.
- **DELETE** is used to remove information.

When a request is sent, the server processes it and sends back a response. The response usually includes a status code, such as **200** if everything worked, and it may also include data that the program can use.

2. JSON and XML

What is JSON?

JSON stands for JavaScript Object Notation. It is a way of storing and sharing data between applications. JSON is popular because it is simple to read and write, and it works well with most programming languages.

Advantages of JSON

- It is easy to read and understand.
- It uses less space because it is lightweight.
- It works well with web APIs.
- Most programming languages support it.

Disadvantages of JSON

- It does not support comments.
 - It is not the best choice for very complicated data.
 - It has fewer built-in validation features than XML.
 - Large JSON files can become difficult to read.
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What is XML?

XML stands for Extensible Markup Language. It is another format used to store and transfer data. XML uses tags to organise information and is often used in larger business systems.

Advantages of XML

- It is good for organising complex data.
- It supports data validation.
- It is supported by many different systems.
- It is useful when data needs a clear structure.

Disadvantages of XML

- It is more difficult to read than JSON.
 - XML files are usually larger.
 - It takes more time to process.
 - Writing XML can be more complicated.
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3. RESTful APIs

What is a RESTful API?

A RESTful API is a way for different applications to communicate with each other over the internet. It allows one application to request information or send information to another application using HTTP requests.

How does a RESTful API work?

A client, such as a website or mobile app, sends a request to a server. The server processes the request and sends back a response. The information is usually returned in JSON format.

For example, an online shopping website can request a list of products from a server. The server gets the product information from the database and sends it back to the website so it can be displayed to the user.

Advantages of RESTful APIs

- They are easy to understand and use.
- They work with many programming languages.
- They are good for building web and mobile applications.
- They make it easier for different systems to share information.

Disadvantages of RESTful APIs

- They need an internet connection to work.
- Too many requests can slow down a server.
- They need proper security to protect data.
- Different API versions can make maintenance more difficult.